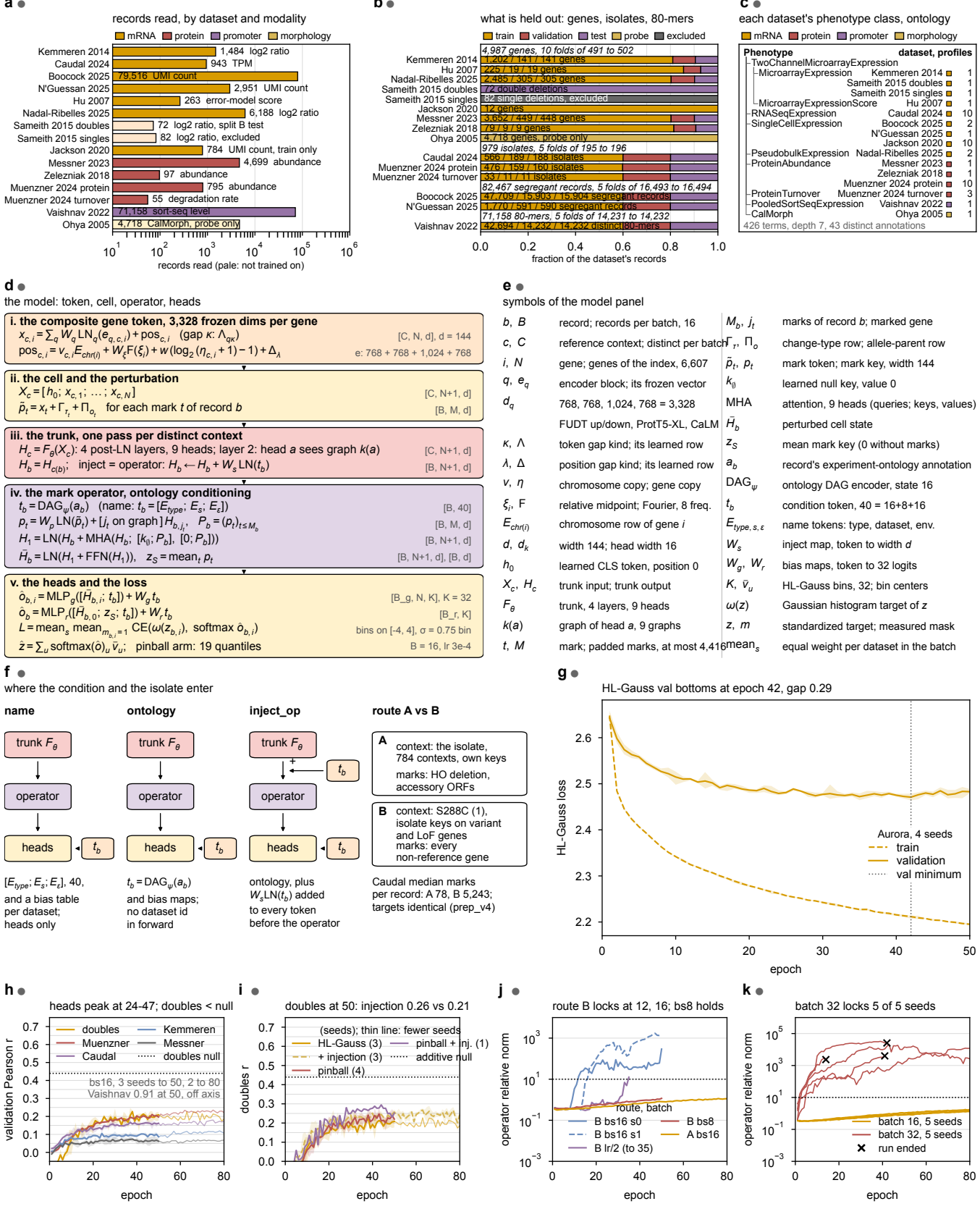

a
records read, by dataset and modality
mRNA
protein
promoter
morphology
Kemmeren 2014
Caudal 2024
Boocock 2025
N'Guessan 2025
Hu 2007
Nadal-Ribelles 2025
Sameith 2015 doubles
Sameith 2015 singles
Jackson 2020
Messner 2023
Zelezniak 2018
Muenzner 2024 protein
Muenzner 2024 turnover
Vaishnav 2022
Ohya 2005
1,484 log2 ratio
943 TPM
79,516 UMI count
2,951 UMI count
263 error-model score
6,188 log2 ratio
72 log2 ratio, split B test
82 log2 ratio, excluded
784 UMI count, train only
4,699 abundance
97 abundance
795 abundance
55 degradation rate
71,158 sort-seq level
4,718 CalMorph, probe only
records read (pale: not trained on)
b
what is held out: genes, isolates, 80-mers
train
validation
test
probe
excluded
4,987 genes, 10 folds of 491 to 502
1,202 / 141 / 141 genes
225 / 19 / 19 genes
2,485 / 305 / 305 genes
72 double deletions
82 single deletions, excluded
12 genes
3,652 / 449 / 448 genes
79 / 9 / 9 genes
4,718 genes, probe only
979 isolates, 5 folds of 195 to 196
566 / 189 / 188 isolates
476 / 159 / 160 isolates
33 / 11 / 11 isolates
82,467 segregant records, 5 folds of 16,493 to 16,494
47,709 / 15,903 / 15,904 segregant records
1,770 / 591 / 590 segregant records
71,158 80-mers, 5 folds of 14,231 to 14,232
42,694 / 14,232 / 14,232 distinct 80-mers
fraction of the dataset's records
c
each dataset's phenotype class, ontology
mRNA
protein
promoter
morphology
Phenotype
dataset, profiles
TwoChannelMicroarrayExpression
MicroarrayExpression
Kemmeren 2014 1
Sameith 2015 doubles 1
Sameith 2015 singles 1
MicroarrayExpressionScore
Hu 2007 1
RNASeqExpression
Caudal 2024 10
SingleCellExpression
Boocock 2025 2
N'Guessan 2025 1
Jackson 2020 10
PseudobulkExpression
Nadal-Ribelles 2025 2
ProteinAbundance
Messner 2023 1
Zelezniak 2018 1
Muenzner 2024 protein 10
ProteinTurnover
Muenzner 2024 turnover 3
PooledSortSeqExpression
Vaishnav 2022 1
CalMorph
Ohya 2005 1
426 terms, depth 7, 43 distinct annotations
d
the model: token, cell, operator, heads
i. the composite gene token, 3,328 frozen dims per gene
$x_{c,i} = \sum_q W_q \mathrm{LN}_q(e_{q,c,i}) + \mathrm{pos}_{c,i}$ (gap κ: $\Lambda_{q\kappa}$)
$\mathrm{pos}_{c,i} = v_{c,i} E_{chr(i)} + W_\xi F(\xi_i) + w(\log_2(\eta_{c,i}+1) - 1) + \Delta_\lambda$
[C, N, d], d = 144
e: 768 + 768 + 1,024 + 768
ii. the cell and the perturbation
$X_c = [h_0; x_{c,1}; \ldots; x_{c,N}]$
$\tilde{p}_t = x_t + \Gamma_{\tau_t} + \Pi_{o_t}$ for each mark t of record b
[C, N+1, d]
[B, M, d]
iii. the trunk, one pass per distinct context
$H_c = F_\theta(X_c)$: 4 post-LN layers, 9 heads; layer 2: head a sees graph $k(a)$
$H_b = H_{c(b)}$; inject = operator: $H_b \leftarrow H_b + W_s \mathrm{LN}(t_b)$
[C, N+1, d]
[B, N+1, d]
iv. the mark operator, ontology conditioning
$t_b = \mathrm{DAG}_\psi(a_b)$ (name: $t_b = [E_{type}; E_s; E_\varepsilon]$)
$p_t = W_p \mathrm{LN}(\tilde{p}_t) + [j_t \text{ on graph}] H_{b,j_t}$, $P_b = (p_t)_{t \le M_b}$
$H_1 = \mathrm{LN}(H_b + \mathrm{MHA}(H_b; [k_\emptyset; P_b], [0; P_b]))$
$\bar{H}_b = \mathrm{LN}(H_1 + \mathrm{FFN}(H_1))$, $z_S = \mathrm{mean}_t\, p_t$
[B, 40]
[B, M, d]
[B, N+1, d]
[B, N+1, d], [B, d]
v. the heads and the loss
$\hat{o}_{b,i} = \mathrm{MLP}_\theta([\bar{H}_{b,i}; t_b]) + W_g t_b$
$\hat{o}_b = \mathrm{MLP}_r([H_{b,0}; z_S; t_b]) + W_r t_b$
$L = \mathrm{mean}_s\, \mathrm{mean}_{m_{b,i}=1}\, \mathrm{CE}(\omega(z_{b,i}),\ \mathrm{softmax}\, \hat{o}_{b,i})$
$\dot{z} = \sum_u \mathrm{softmax}(\hat{o})_u \bar{v}_u$; pinball arm: 19 quantiles
[B_g, N, K], K = 32
[B_r, K]
bins on [-4, 4], σ = 0.75 bin
B = 16, lr 3e-4
e
symbols of the model panel
b, B record; records per batch, 16
c, C reference context; distinct per batch
i, N gene; genes of the index, 6,607
q, e_q encoder block; its frozen vector
d_q 768, 768, 1,024, 768 = 3,328
FUDT up/down, ProtT5-XL, CaLM
κ, Λ token gap kind; its learned row
λ, Δ position gap kind; its learned row
v, η chromosome copy; gene copy
ξ_i, F relative midpoint; Fourier, 8 freq.
$E_{chr(i)}$ chromosome row of gene i
d, d_k width 144; head width 16
h_0 learned CLS token, position 0
X_c, H_c trunk input; trunk output
F_θ trunk, 4 layers, 9 heads
$k(a)$ graph of head a, 9 graphs
t, M mark; padded marks, at most 4,416
M_b, j_t marks of record b; marked gene
Γ_τ, Π_o change-type row; allele-parent row
$\tilde{p}_t$, p_t mark token; mark key, width 144
$k_\emptyset$ learned null key, value 0
MHA attention, 9 heads (queries; keys, values)
$\bar{H}_b$ perturbed cell state
z_S mean mark key (0 without marks)
a_b record's experiment-ontology annotation
DAG_ψ ontology DAG encoder, state 16
t_b condition token, 40 = 16+8+16
$E_{type,s,\varepsilon}$ name tokens: type, dataset, env.
W_s inject map, token to width d
W_g, W_r bias maps, token to 32 logits
K, $\bar{v}_u$ HL-Gauss bins, 32; bin centers
$\omega(z)$ Gaussian histogram target of z
z, m standardized target; measured mask
mean_s equal weight per dataset in the batch
f
where the condition and the isolate enter
name
ontology
inject_op
route A vs B
trunk F_θ
operator
heads
t_b
$[E_{type}; E_s; E_\varepsilon]$, 40,
and a bias table
per dataset;
heads only
$t_b = \mathrm{DAG}_\psi(a_b)$
and bias maps;
no dataset id
in forward
ontology, plus
$W_s \mathrm{LN}(t_b)$ added
to every token
before the operator
A context: the isolate, 784 contexts, own keys
marks: HO deletion, accessory ORFs
B context: S288C (1), isolate keys on variant and LoF genes
marks: every non-reference gene
Caudal median marks
per record: A 78, B 5,243;
targets identical (prep_v4)
g
HL-Gauss val bottoms at epoch 42, gap 0.29
HL-Gauss loss
epoch
Aurora, 4 seeds
train
validation
val minimum
h
heads peak at 24-47; doubles < null
validation Pearson r
epoch
doubles
Kemmeren
Muenzner
Messner
Caudal
doubles null
bs16, 3 seeds to 50, 2 to 80
Vaishnav 0.91 at 50, off axis
i
doubles at 50: injection 0.26 vs 0.21
doubles r
epoch
(seeds); thin line: fewer seeds
HL-Gauss (3)
pinball + inj. (1)
+ injection (3)
additive null
pinball (4)
j
route B locks at 12, 16; bs8 holds
operator relative norm
epoch
route, batch
B bs16 s0
B bs8
B bs16 s1
A bs16
B lr/2 (to 35)
k
batch 32 locks 5 of 5 seeds
operator relative norm
epoch
batch 16, 5 seeds
batch 32, 5 seeds
run ended